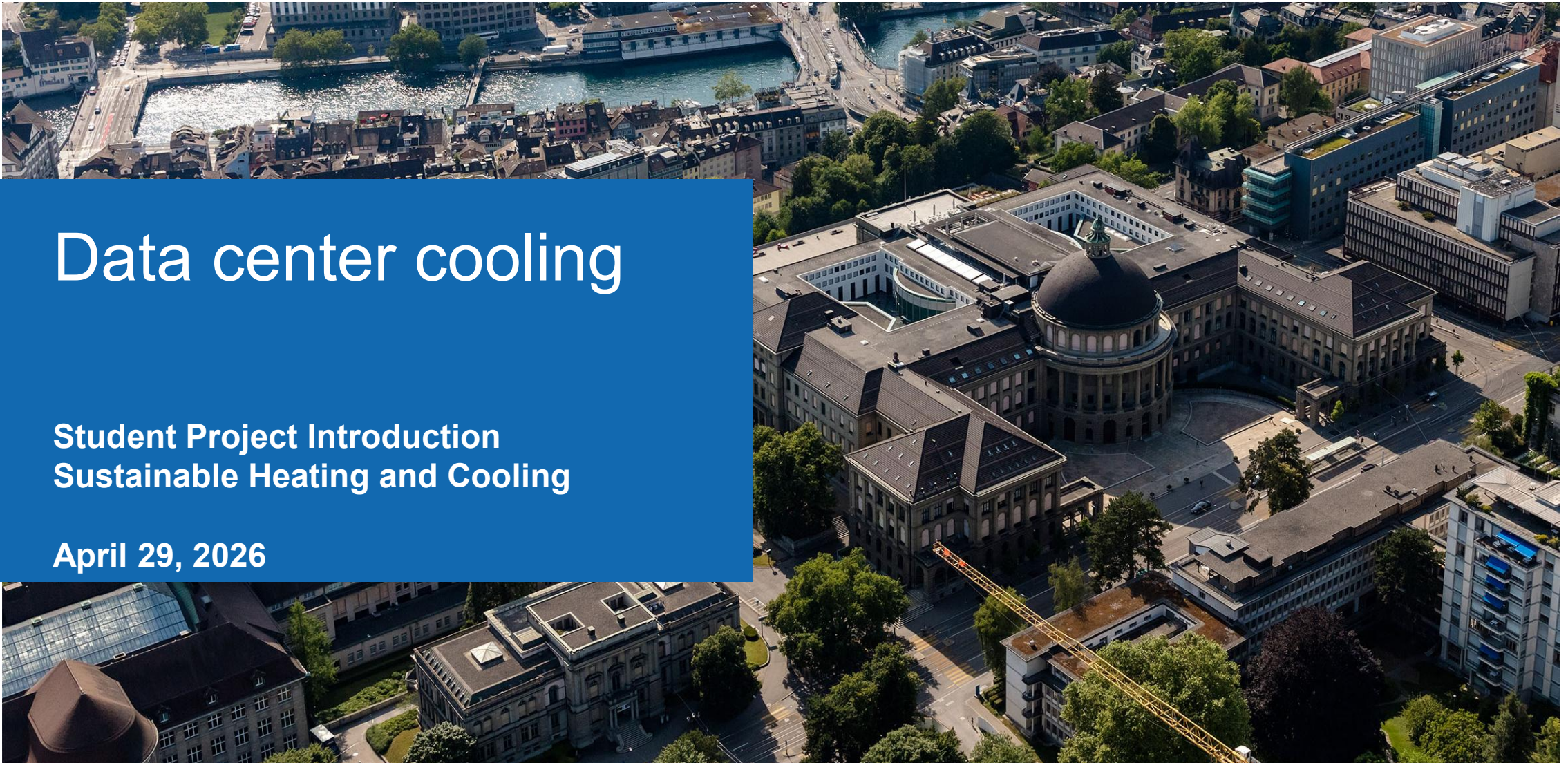


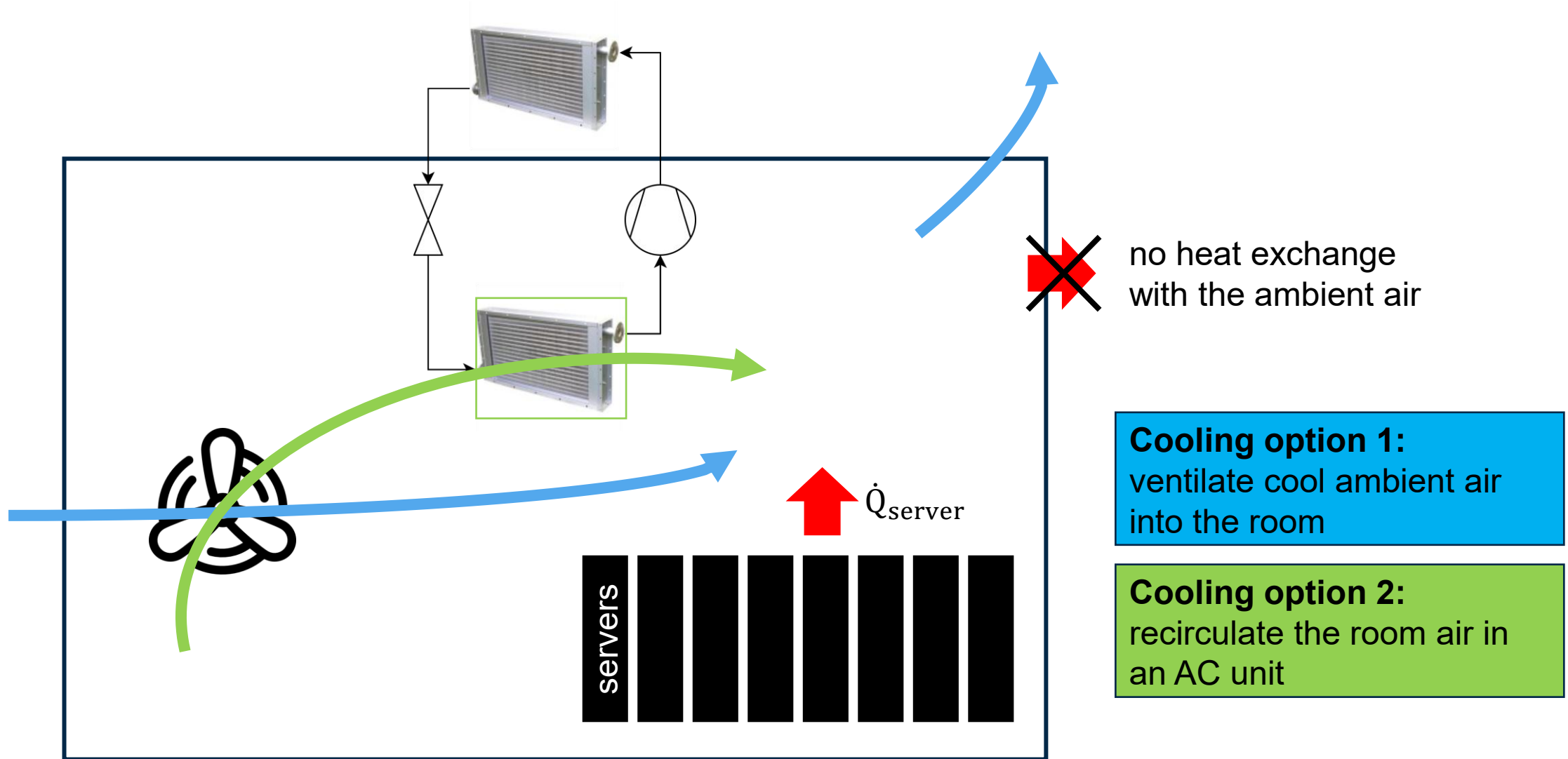
Data center cooling

Student Project Introduction
Sustainable Heating and Cooling

April 29, 2026



System description



Your tasks

1. System Requirements

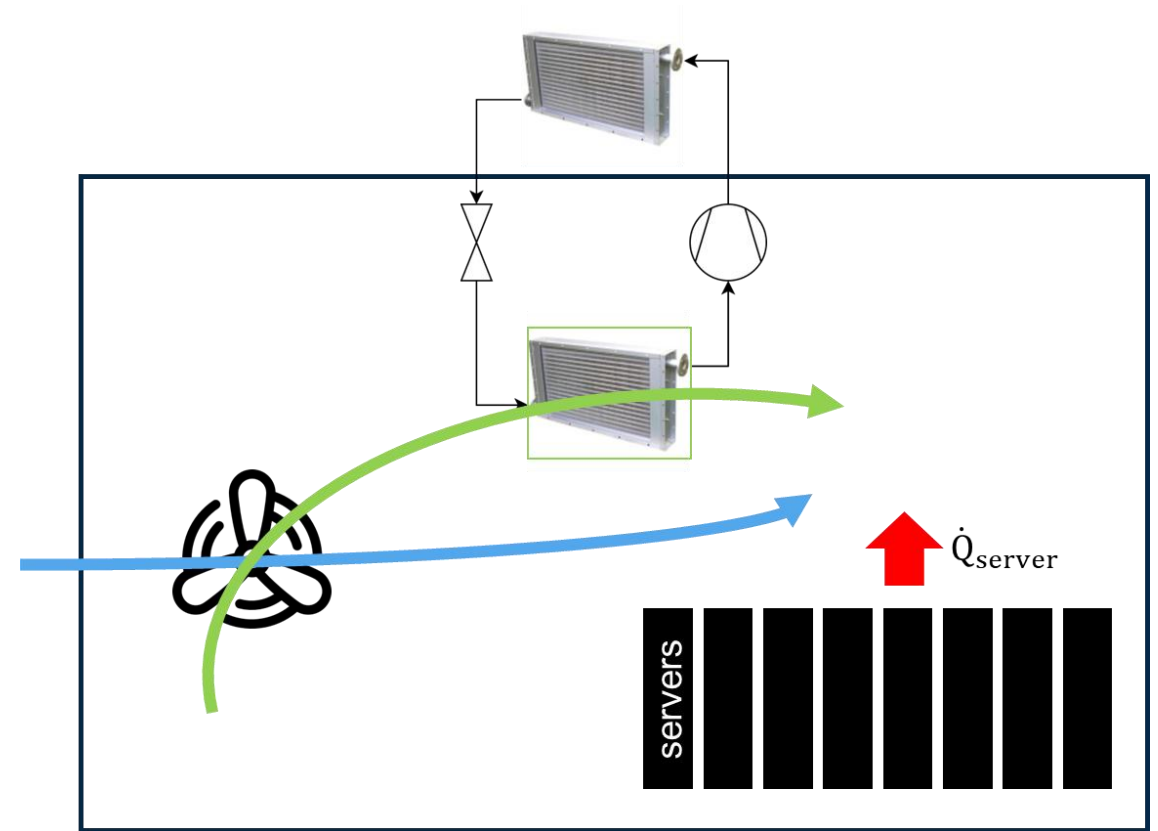
Determine system constraints and find a control strategy

2. Design

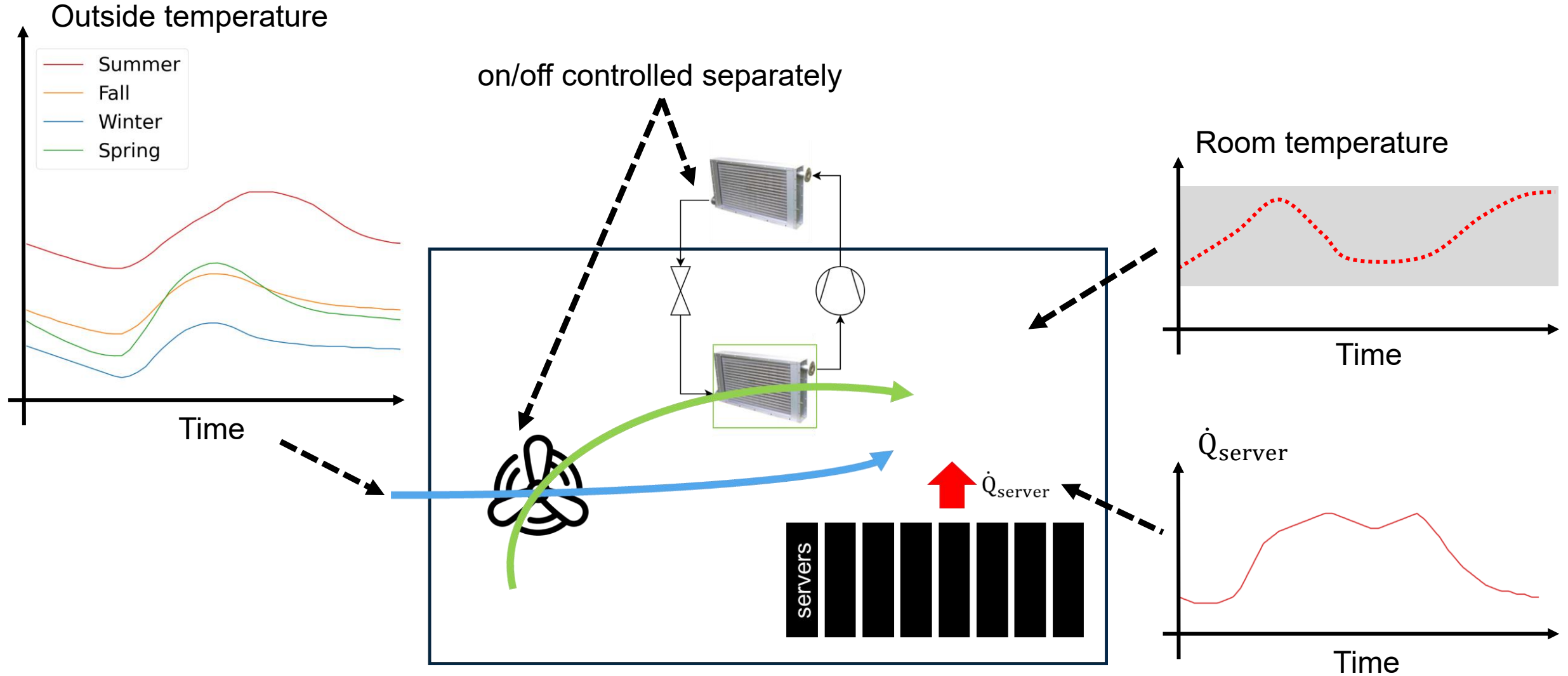
Consider equipment sizes and the refrigerant to design the ventilation and AC system

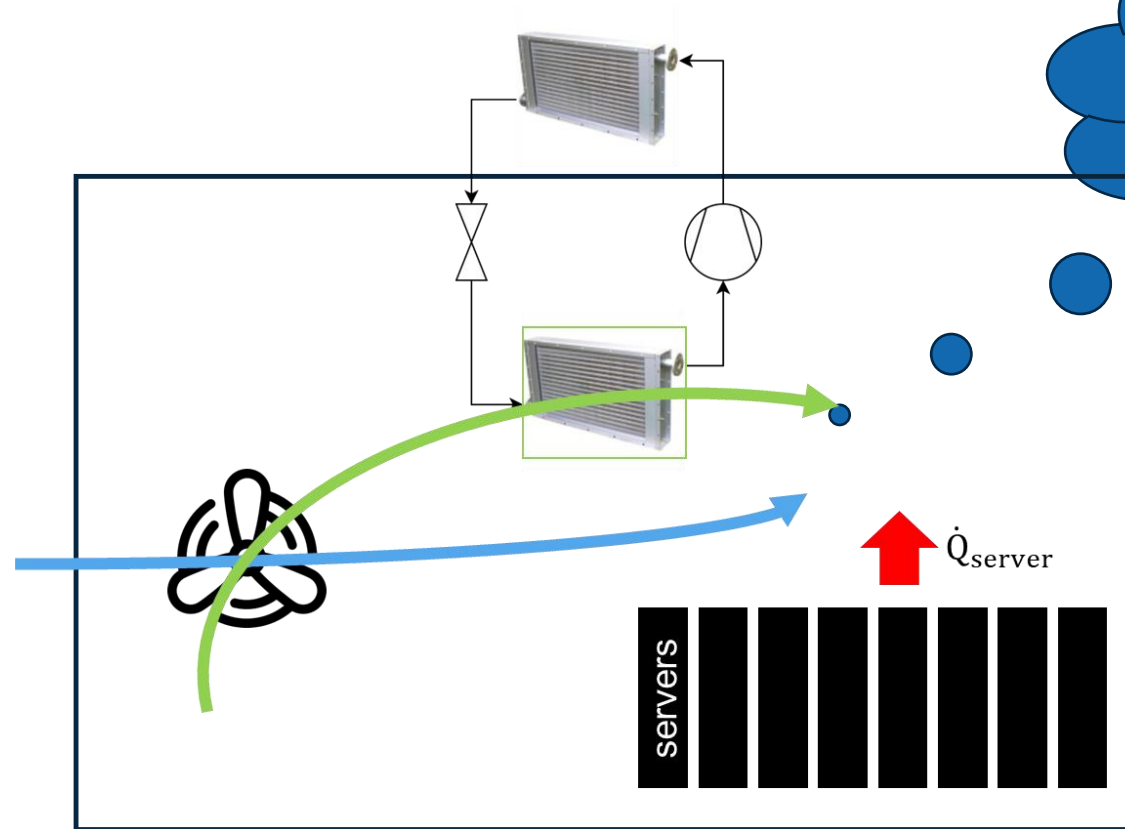
3. Simulation

Simulate the operation and iterate your design



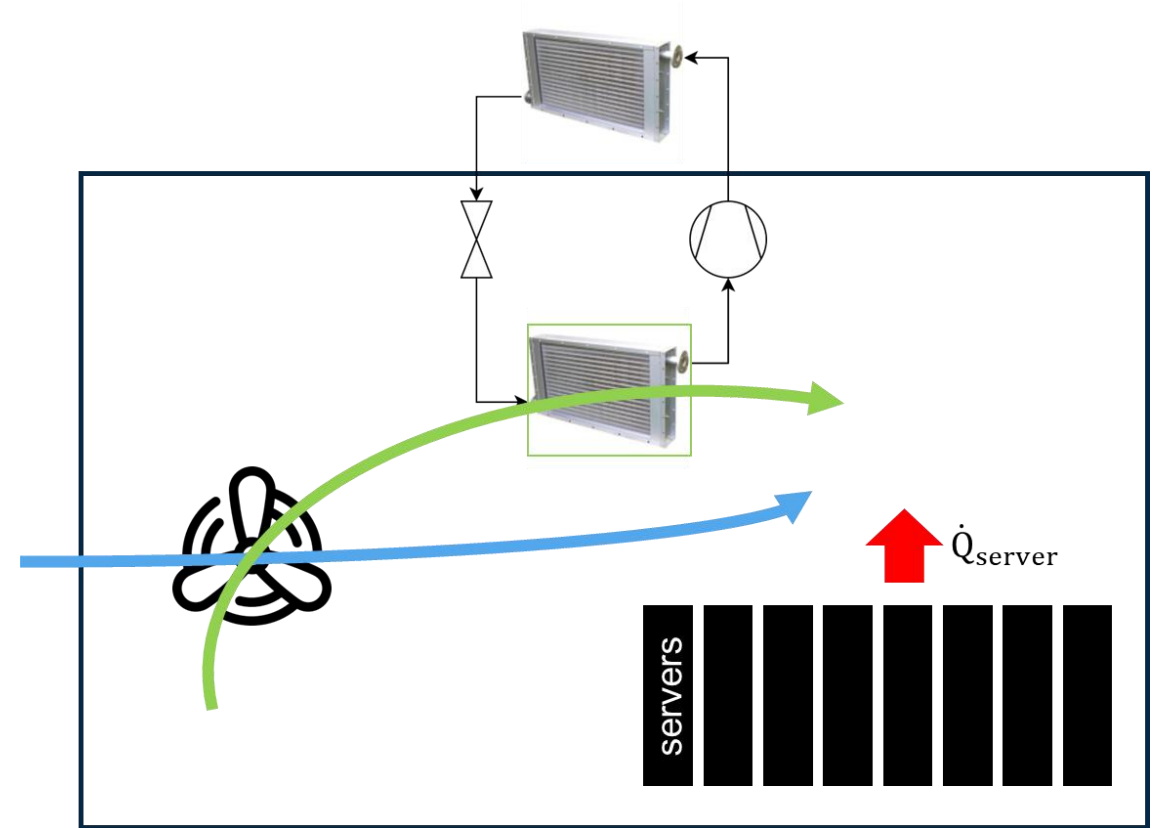
System Requirements



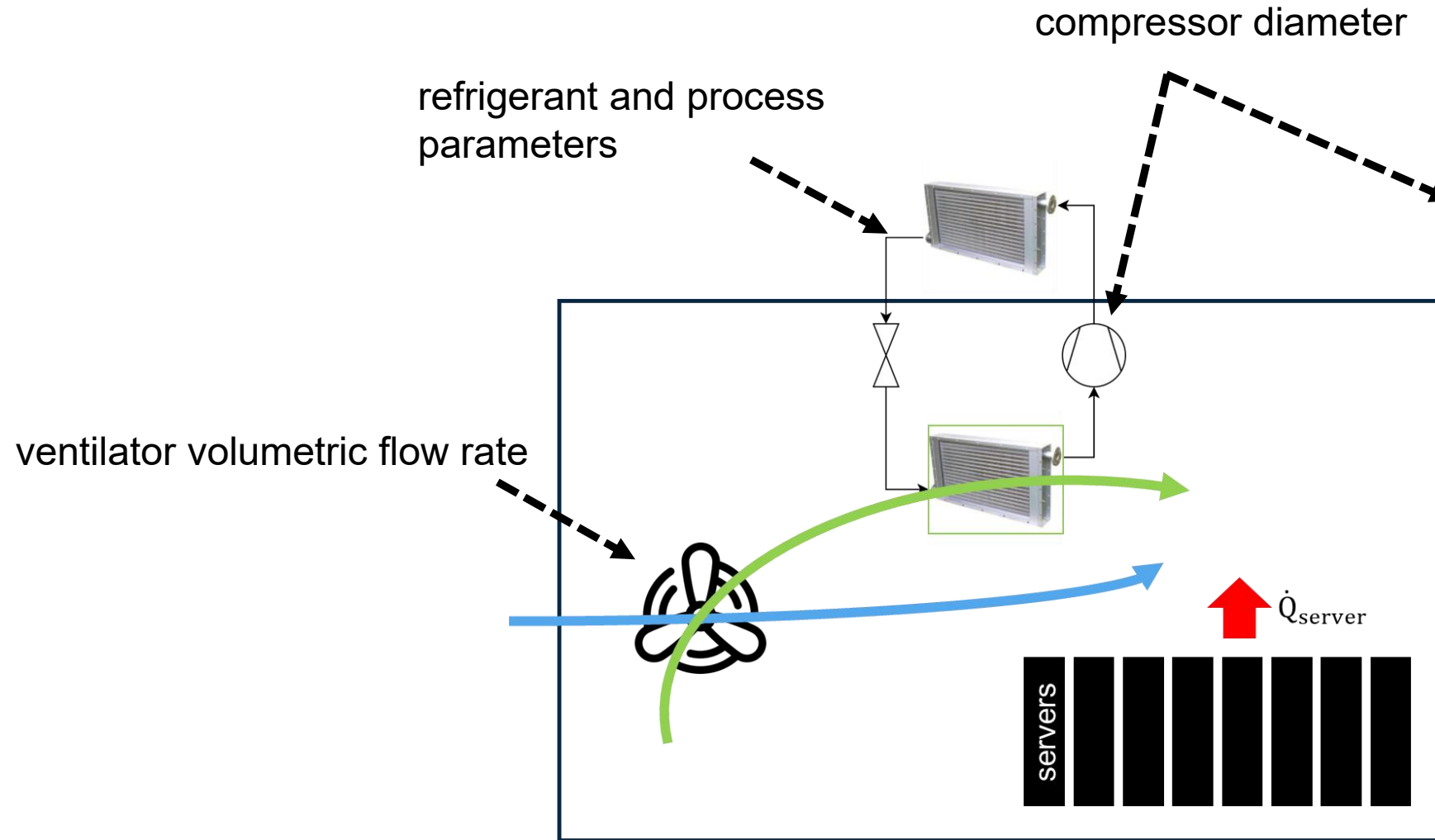


When do we run the ventilation and when the AC?

- Take the system requirements and constraints into account
- Design the cooling system such that the room temperature stays within the bounds



Design variables



`recip_comp_corr_SP(param, refrigerant)`



returns η_{is} , $\dot{m}_{\text{dot}}$

Design variables

	D = 30 mm	D = 40 mm	D = 50 mm
Propane			
Dimethyl ether			
R1234yf			

`recip_comp_corr_SP(param, refrigerant)`



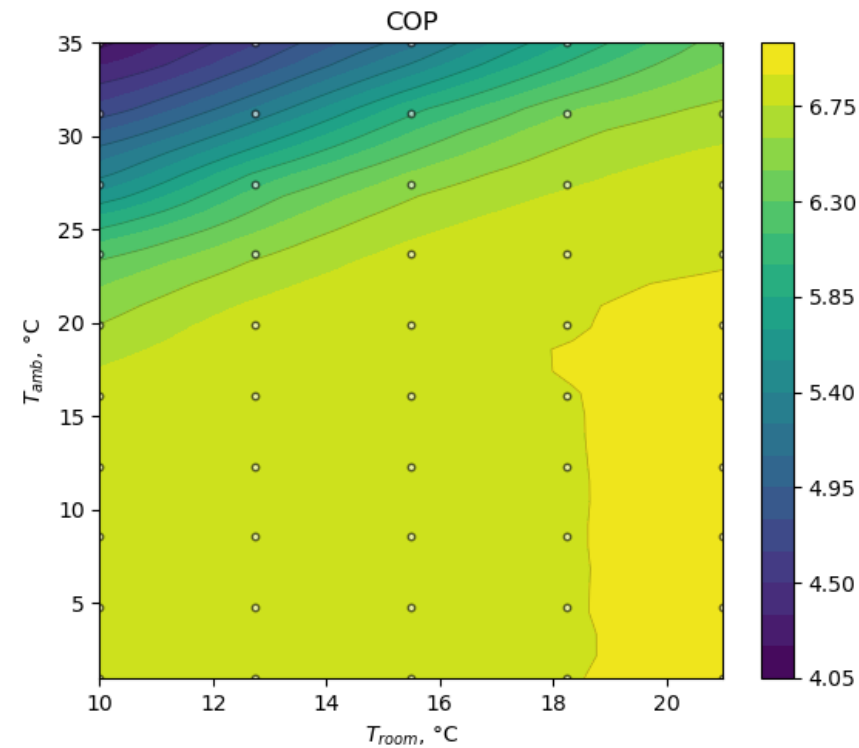
returns `eta_is`, `m_dot`

- 4 cylinders
- Available in 3 sizes

Simulate the system performance over a full day for each of the 4 seasons to determine your best system design and operation.

Explain your selection criteria and discuss your decision!

To avoid repeated cycle optimization during simulation, we recommend to determine the COP and cooling capacity as functions of the source and sink temperatures!



- Groups have been assigned → Contact us if there are any issues!
- Report
 - You may use the provided template (additional information)
 - Completeness over length! (max. 10 pages worth of text)
 - Clean figures, clear and readable labels
- Office hour: May 27, 02:15 – 03:45pm, HG E 19
(2nd office hour tba)
- Submission via Moodle course: June 28

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